Uncertainty-Guided Pseudo-Label Refinement for Long-Tail 3D Scenes

B.Tech in Computer Science and Engineering

Ananya Srivastava

122CS0077

Agenda

- Introduction
- Literature Review
- Research Gaps Identified
- Problem Statement
- Objectives
- Proposed Methodology
- Implementation Details
- Results
- Conclusion
- References

Introduction

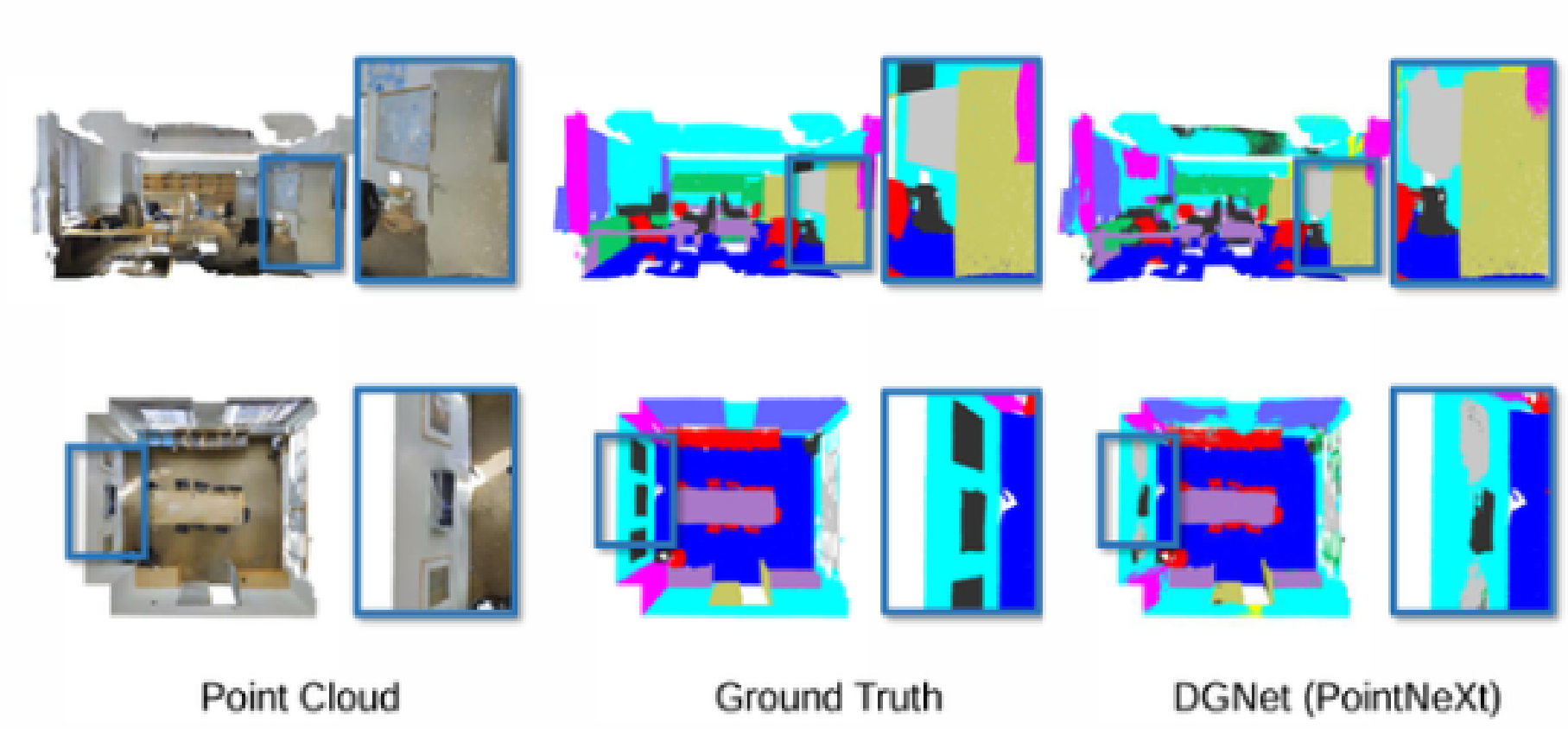

Figure: Visual comparisons between Ground Truth and DGNet Semantic segmentation on S3DIS Area 5 at 0.01/% label rate.

Introduction & Motivation

The Demand for 3D Perception:

- 3D point cloud segmentation is critical for autonomous systems.
- Let a point cloud be $\mathcal{P} = \{p_i \in \mathbb{R}^3\}_{i=1}^N$. The goal is to learn a mapping $f_\theta : \mathcal{P} \to \mathbb{R}^{N \times C}$ for C classes.

The Annotation Bottleneck:

- Dense, point-level annotation of large-scale 3D scenes is prohibitively expensive.

The Role of Pseudo-Labeling (PL):

- To utilize unlabeled data $\mathcal{P}^U$, Pseudo-Label Generators (PLGs) assign labels based on model confidence:

$$\hat{y}_i = \arg\max_c p(y_i = c \mid p_i; \theta) \quad \text{if} \quad \max_c p(y_i = c \mid p_i; \theta) > \tau$$

- The quality of $\hat{y}_i$ dictates the absolute performance ceiling of the model.

Literature Review

Paper	Year/ Venue	Key Points	Gaps	Dataset	Model
Pseudo-labeling and confirmation bias in deep semi-supervised learning	2020 arXiv	Generates soft pseudo-labels directly from predictions; utilizes mixup to reduce confirmation bias.	Naive pseudo-labeling overfits to own errors; consistency needs heavy augmentation.	CIFAR, SVHN, Mini-ImageNet	Soft pseudo-labeling with mixup and sample over-sampling.
In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning	2021 ICLR	Proposes Uncertainty-aware Pseudo-label Selection (UPS); creates negative labels for multi-label tasks.	Poor network calibration produces confident but incorrect pseudo-labels.	CIFAR, Pascal VOC	UPS framework.

Literature Review

Paper	Year/ Venue	Key Points	Gaps	Dataset	Model
Domain-generalizable PC inst. seg. of bridge cmpts. using class-balanced dynamic thresholding	2026 Aut. in Const.	Introduces unsupervised domain adaptation and Class-Balanced Dynamic Thresholding (CBDT).	Detailed 3D labels are costly; severe domain shifts and long-tail class imbalance.	Synthetic & Real bridges	SoftGroup++ combined with CBDT.
Distribution guidance network for weakly supervised pc se. seg.	2024 NeurIPS	Regulates feature space using moVMF distributions; dynamically aligns network embeddings to distribution priors.	Weak supervision lacks signals; ignores inherent distribution of embeddings, causing ambiguity.	S3DIS, ScanNetV2	Distribution Guidance Network (DGNet).

Literature Review

Paper	Year/ Venue	Key Points	Gaps	Dataset	Model
Joint global and dynamic pseudo labeling for semi-supervised point cloud sequence segmentation	2023 IEEE TCSVT	Generates global and dynamic pseudo-labels via K-NN search; merges them for sequence segmentation.	3D point-to-point tracking is hard; existing SSL ignores sequence dynamics.	SemanticKITTI, Synthia	GTD+, Pseudo label generation with SPVCNN.
Pseudo Label Learning for Partial Point Cloud Registration	2025 IEEE TVCG	Generates pseudo-labels for overlap regions via nearest points in aligned cloud pairs.	Real point clouds have low overlap/missing data; heavy reliance on ground truth labels.	ModelNet40, KITTI	PSEL framework with dynamic AGCNN.

Literature Review

Paper	Year/ Venue	Key Points	Gaps	Dataset	Model
SeCoV2: Semantic Connectivity-Driven Pseudo-Labeling for Robust Cross-Domain Semantic Segmentation	2025 IEEE TPAMI	Refines pseudo-labels at the connectivity level; groups pixels using 2D foundation models & graph correction.	Fragmented pixel predictions under domain shift; SAM merges unrelated objects without semantics.	GTA5, Cityscapes	SeCoV2 (PSA + SCC-Unc modules).
Weakly supervised point cloud semantic segmentation via artificial oracle	2024 CVPR	Uses 2D foundation models (SAM) as an artificial oracle; samples confident queries without human annotators.	Active learning requires a human-in-the-loop; applying 2D SAM directly to 3D lacks semantics.	S3DIS, ScanNetV2	REAL framework with PP2S projection module.

Literature Review

Paper	Year/ Venue	Key Points	Gaps	Dataset	Model
UniLiPs: Unified LiDAR Pseudo-Labeling with Geometry-Grounded Dynamic Scene Decomposition	2026 arXiv	Unsupervised multi-modal labeling; uses 2D foundation models to generate 3D geometry priors dynamically.	3D LiDAR annotation is a major bottleneck; usually requires manual input for separate tasks.	KITTI, nuScenes	UniLiPs pipeline with iterative 2D-3D fusion.
Segment everything everywhere all at once	2023 NeurIPS	Unifies semantic/instance segmentation; uses varied prompts (text, box, point) mapped to one visual-semantic space.	Lack of a universal interface handling multiple prompts; standard SAM lacks open-vocabulary semantics.	COCO, LVIS, PascalVOC	SEEM (Transformer encoder-decoder).

Research Gaps Identified

Confirmation Bias in PLGs:

- Models become overconfident in incorrect predictions. Minimizing standard cross-entropy on noisy pseudo-labels $\hat{y}_i$ reinforces errors.

Unresolved Long-Tail Imbalance:

- 3D data exhibits severe imbalance: $P(C_{head}) \gg P(C_{tail})$.
- PLGs naturally bias towards the majority class prior, systematically ignoring rare objects like pedestrians or traffic signs.

Wasted "Hard Prompts":

- Let $U(p_i)$ be the uncertainty of point p_i. Current frameworks can locate highly uncertain points ($U(p_i) > \tau_{unc}$), but standard PLGs discard them, wasting valuable minority-class data.

Problem Statement

The Core Issue: Current PLGs favor common classes, leading to a bias where models forget or misclassify rare 'tail' objects.

The Bottleneck: While uncertainty-guided mining identifies tricky regions, standard PLGs lack a mechanism to convert these "hard prompts" into reliable supervision.

The Impact: This generates noisy pseudo-labels $\tilde{y}_i$, which disrupt training and severely skew the learned distribution $\tilde{P}_\theta(Y)$ away from the true underlying distribution $P(Y)$.

Proposed Objectives

1. **Uncertainty-Guided Mining:**
   Implement an isolation mechanism to identify "hard prompts" representing tail classes, defined mathematically as the subset:

$$\mathcal{P}_{hard} = \{p_i \in \mathcal{P}^U \mid U(p_i) > \tau_{unc}\}$$

2. **Dynamic Label Refinement:**
   Develop a correction module Φ (utilizing geometric priors or 2D oracles, $\mathcal{O}$) that maps uncertain predictions into reliable supervision:

$$y_i^{ref} = \Phi(\tilde{y}_i, p_i, \mathcal{O}), \quad \forall p_i \in \mathcal{P}_{hard}$$

3. **Framework Evaluation:**
   Optimize the model using the refined labels to demonstrate a reduction in confirmation bias and measurable mIoU improvements specifically on C_{tail} categories.

Methodology

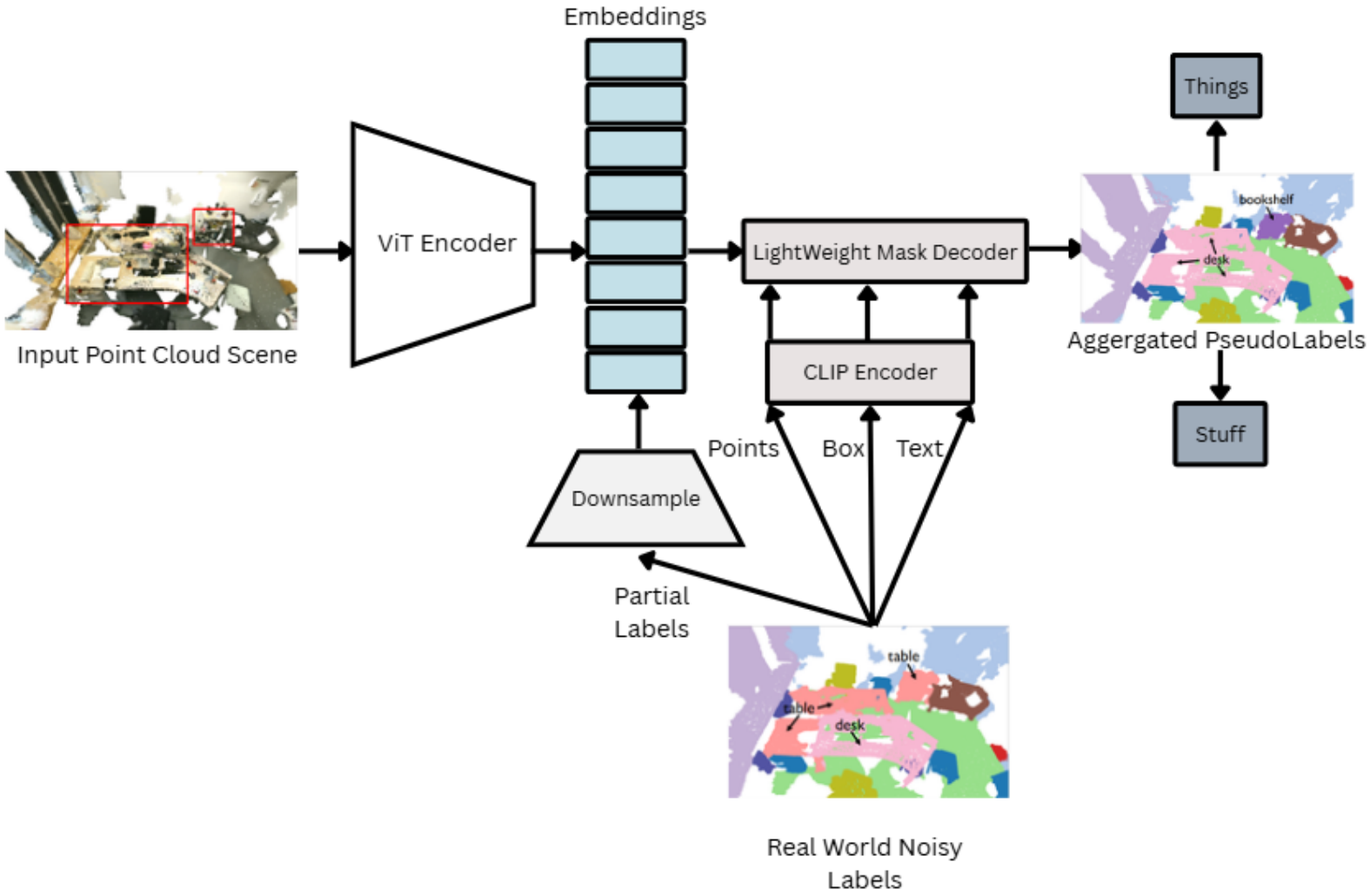

Figure: Aggeragation of Point level pseudolabels for a given 3D scene

Proposed Methodology

The Connectivity Paradigm Shift: Instead of optimizing isolated, noisy point/pixel predictions under severe domain shifts, we reformulate pseudo-labeling at the *structural connectivity level*.

Phase 1: Point-Prompt Semantic Aggregation
We group initial predictions into coherent connectivities utilizing a foundation model (SAM) via a **Dual-Prompting Strategy**:

- **Things Categories:** Use extracted bounding boxes and centroids as explicit prompts to preserve fine-grained instance boundaries.
- **Stuff Categories:** Use uniform grid prompts to generate class-agnostic masks, assigning labels via *majority voting* of the covered pixels.

Methodology

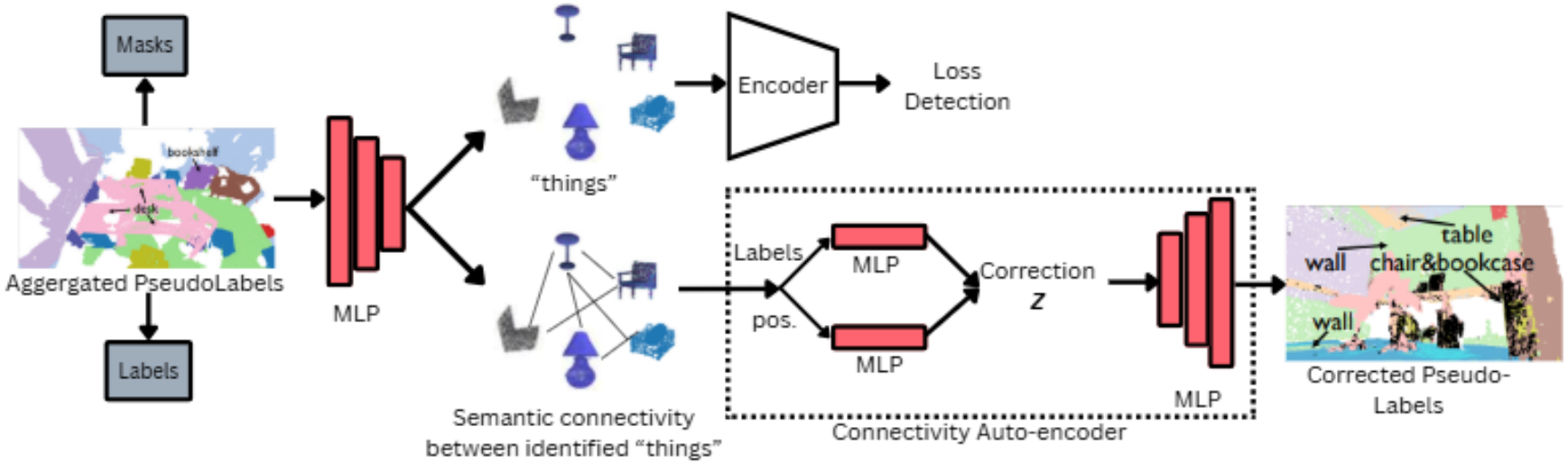

Figure: Depiction of Semantic Connectivity Correction and Self training Phase

Methodology

Phase 2: Semantic Connectivity Correction

To identify and filter semantic noise within the aggregated connectivities, we leverage learning dynamics:

- **Early Stopped Classifier:** A lightweight classifier is trained briefly on the connectivities.
- **Loss Distribution Modeling:** Clean samples naturally exhibit lower loss, while noisy samples exhibit high loss.
- **Filtering:** We fit a distribution model to separate the connectivities into a clean subset ($\mathcal{D}_{clean}$) and a corrected subset ($\mathcal{D}_{corr}$).

Phase 3: Iterative Self-Training

The robust, filtered connectivities ($\mathcal{D}_{clean} \cup \mathcal{D}_{corr}$) provide high-quality supervision for the unlabeled target domain. The main segmentation network is then jointly optimized across both domains.

Dataset Analysis: The Long-Tail Problem

Table: Detailed profile of class imbalance across evaluated datasets.

Dataset	**Scale & Classes**	**Typical Tail Classes**	**Root Cause of Imbalance**
SemanticKITTI (Outdoor Driving)	**Classes:** 19 **Scale:** 43,552 LiDAR scans (∼4.5 Billion points)	Bicyclist, Motorcyclist, Person, Traffic-sign	Spatial dominance
nuScenes (Urban Scenes)	**Classes:** 16 **Scale:** 1,000 scenes (∼1.4 Billion points)	Bicycle, Motorcycle, Pedestrian, Traffic Cone	Sensor sparsity & thin structures
ScanNetV2 (Indoor Spaces)	**Classes:** 20 **Scale:** 1,513 room scenes	Picture, Toilet, Sink, Shower Curtain	Room-specificity & ccclusion

Implementation Details: Datasets & Metrics

Datasets Evaluated:

- **SemanticKITTI:** Large-scale outdoor autonomous driving dataset. Highly imbalanced class distribution.
- **nuScenes:** Diverse urban outdoor scenes with a sharp long-tail drop-off.
- **ScanNetV2:** Indoor cluttered spaces, with heavy occlusion and ambiguous boundaries.

Evaluation Metrics:

- **Overall mIoU (%):** Standard metric for general segmentation accuracy.
- **Tail-class mIoU (%):** Crucial metric isolating the performance on the most underrepresented minority categories (e.g., bicycles, signs).
- **Expected Calibration Error (ECE %):** Measures the gap between model confidence and actual accuracy. A lower ECE indicates a reduction in confirmation bias.

Implementation Details: Environment & Baselines

Hardware Environment:

- All model benchmarking and iterative refinement were executed utilizing a **Kaggle Tesla P100 GPU**.
- *Note:* Demonstrates the efficiency of the chosen architectures under constrained computational resources.

Models Evaluated:

1. **SPVCNN:** The standard, naive pseudo-labeling baseline.
2. **DGNET:** Represents mathematical prior alignment (moVMF distributions).
3. **GD-Pseudo:** Represents geometry-driven, continuous surface propagation.
4. **REAL:** Represents 2D Foundation Model (SAM) active oracles.
5. **UniLiPs:** The State-of-the-Art multi-modal fusion baseline.

Quantitative Results: Benchmarking

Table: Performance comparison across datasets for 3D Long-Tail Segmentation

2*Model	SemanticKITTI			nuScenes			ScanNetV2		
	mIoU	**Tail-mIoU**	**ECE**	**mIoU**	**Tail-mIoU**	**ECE**	**mIoU**	**Tail-mIoU**	**ECE**
SPVCNN	52.03	14.94	15.07	55.19	18.77	14.57	51.30	19.39	16.36
DGNET	58.73	25.46	7.13	59.93	27.01	7.21	58.69	26.67	8.60
GD-Pseudo	57.60	22.91	10.19	60.35	25.40	9.00	56.33	22.12	11.98
REAL	62.83	39.37	8.47	64.01	41.28	7.61	65.86	46.79	7.10
UniLiPs	**68.79**	**55.20**	**4.05**	**71.64**	**59.58**	**3.50**	n/a	n/a	n/a

Conclusion: Analyzing the Baselines

The Reality of Confirmation Bias:

- **SPVCNN** exhibits the highest ECE ($\sim$15-16%) and lowest tail-mIoU. Dominant classes overwhelm the pseudo-labels, proving that naive models become highly overconfident in wrong predictions.

Mathematical vs. Geometric Priors:

- **DGNET** successfully slashes ECE (down to $\sim$7%) by explicitly modeling uncertainty via moVMF distributions, rescuing some tail-class gradients.
- **GD-Pseudo** effectively propagates labels along continuous surfaces (roads/walls) but struggles with isolated tail-class objects (e.g., bicycles), yielding moderate tail-mIoU improvements.

Conclusion: The Power of Foundation Oracles

The Shift to 2D-to-3D Oracles:

- **REAL** achieves a massive boost in tail-class mIoU (e.g., 46.79% on ScanNetV2). By leveraging 2D foundation models (SAM), it easily isolates rare objects, proving highly effective in cluttered indoor environments.

State-of-the-Art Synthesis:

- **UniLiPs** operates at near-oracle performance ($\sim$1% gap to fully supervised models).
- By fusing foundation models (SAM2, CLIP) with temporal LiDAR accumulation, it drastically minimizes ECE ($\sim$3.5-4%) and maximizes tail-class accuracy.
- **Final Verdict:** The data confirms that intercepting "hard prompts" with robust geometric and multi-modal priors is the definitive solution to long-tail confirmation bias.

References